DNA Sequence Visualization With Arithmetic Coding

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Abstract

We propose a novel two-dimensional graphical visualization method to intuitively analyze and compare multiple DNA sequences. To do so, we employ arithmetic coding, a lossless entropy-based compression method, to transform a sequence into a representative series of bits, taking advantage of differing DNA k-mer frequencies to encode DNA sequences more efficiently than standard two-bit encoding methods. To better visualize and compare related sequences, the k-mer frequencies used to generate the encoding can be derived from multiple sequences. The compressed binary sequence is then represented by a series of vectors constituting a two-dimensional graphical visualization that avoids degeneracy and loss of information. This DNA visualization method can be used to explore variation across sequences in an efficient and intuitive manner.

Background

With the rise of genome sequencing, DNA sequence analysis has emerged as a major area in bioinformatics, and the development of analytical tools to provide intuitive and meaningful insights into patterns and variations in DNA has become increasingly desirable.

There exist many methods to produce a two-dimensional graphical visualization of DNA sequences. Gates [1] and Nandy [2] propose the use of four directional unit vectors to represent each nucleotide, though both visualizations have degeneracy. Yau et al. [3] assign two upward unit vectors for the pyrimidines C and T and two downward unit vectors for the purines A and G. Randic et al. [4] propose a visualization with four variable positions on the y-axis representing the nucleotides; however, this visualization is difficult to interpret as sequences become longer.

Qi and Qi [5] employ Huffman coding to assign each nucleotide to a binary code, transforming the nucleotide sequence into a series of bits and mapping each bit to a vector. They are the first to propose entropy coding methods as a means to encode DNA sequences prior to visualization. In an orthogonal approach to Qi and Qi, we introduce the use of arithmetic coding to encode nucleotide sequences and generate graphical visualizations of DNA sequences.

Methods

The raw DNA sequence undergoes several transformations prior to visualization. First, the nucleotide sequence is inputted into the visualization algorithm. The incidence of each *k*-mer (Fig. 1) is counted with overlap and a frequency table is generated. This frequency table can be assembled across several related sequences for enhanced comparison.

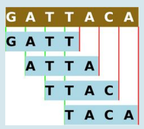


Fig. 1: 4-mers of nucleotide sequence GATTACA.

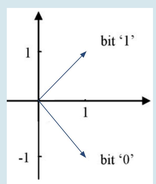
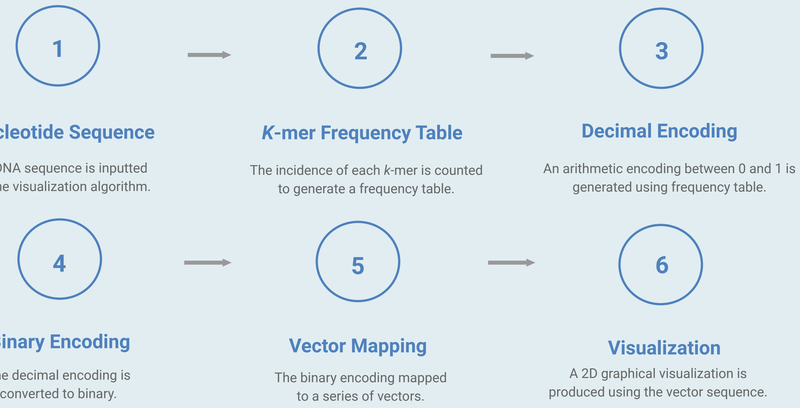


Fig. 2: Bit to vector mapping scheme.

The nucleotide sequence is then encoded arithmetically and an encoded value between 0 and 1 is generated. The decimal encoding is converted into a binary arithmetic encoding. This binary sequence is then mapped to a multicomponent vector using a dual vector mapping method employed by Qi and Qi, as well as Lee [6] and Zhang [7] with (1, 1) corresponding to 1 and (1, -1) corresponding to 0 (Fig. 2).

Workflow



DNA Visualization Graphs

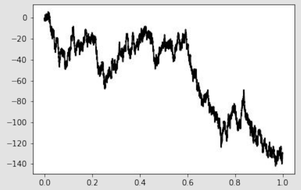


Fig. 3a: Bat HBB gene

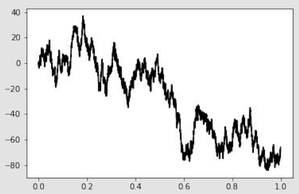


Fig. 3b: Chinchilla HBB gene

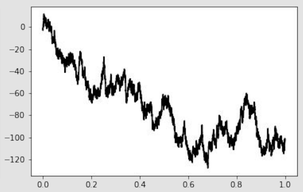


Fig. 3c: Cow HBB gene

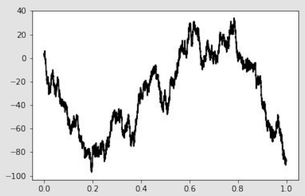


Fig. 3d: Dog HBB gene

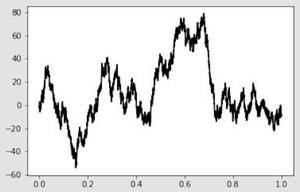


Fig. 3e: Goat HBB gene

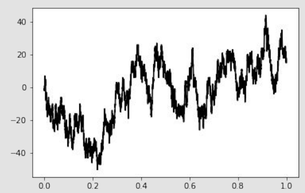


Fig. 3f: Human HBB gene

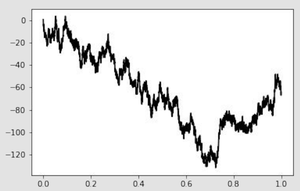


Fig. 3g: Norway Rat HBB gene

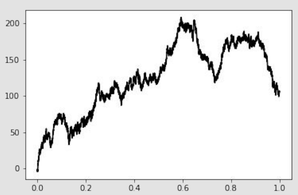


Fig. 3h: Pelican HBB gene

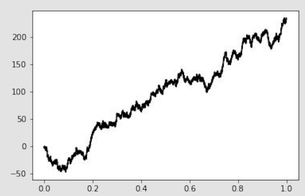


Fig. 3i: Pig HBB gene

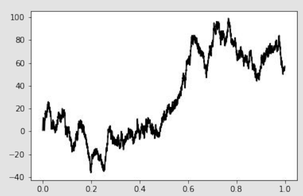


Fig. 3j: Pike HBB gene

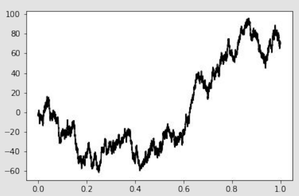


Fig. 3k: Polar Bear HBB gene

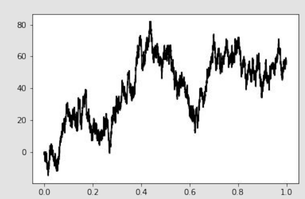


Fig. 3l: Rhesus Monkey HBB gene

Arithmetic Coding with *K*-mers

Input sequence: AAT GGC GGC

Output encoding: 0.07067136

Discussion

The implications of our novel arithmetic encoded DNA visualization method (see Figs. 3a-l) are manifold. Certain existing visualization methods, such as Squiggle [6] and DV-Curve [7], use a two-bit encoding for each nucleotide. However, as arithmetic coding is an entropy coding method, it utilizes repetition of *k*-mers to minimize entropy and allocate fewer than two bits per nucleotide on average, producing a more efficient encoding. If all *k*-mers within a sequence occur at the same frequency, unlikely especially as *k* increases in size, entropy is maximized and a two-bit encoding is produced. Thus, our visualization method requires significantly less information to encode each DNA sequence, which is ideal for longer sequences.

In addition, our algorithm, by virtue of the arithmetic coding method, yields a one-to-one relationship between a unique DNA sequence and its corresponding visualization, preventing degeneracy and loss of information. In addition, *k*-mer frequency tables can be generated across several related sequences, allowing for unique visualizations and more robust comparisons between sequences.

Future Work: We aim to continue to evaluate the results of our DNA visualization algorithm, focusing on longer DNA sequences and entire genomes and improving interpretability. This algorithm can be implemented in a DNA visualization web tool to increase accessibility and ease of use.

References

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Acknowledgements

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